Inhibition of local estrogen synthesis disrupts neuronal and glial homeostasis in adult retinal explants

Purpose

The retina is increasingly recognized as a neuroendocrine tissue capable of locally synthesizing steroid hormones that regulate neuronal survival, glial function, and tissue homeostasis. Estrogens, particularly 17β-estradiol, exert well-established neuroprotective effects in the central nervous system; however, their functional relevance when synthesized locally within the adult retina remains poorly defined. Although altered hormonal signaling has been associated with retinal neurodegeneration, direct experimental evidence linking endogenous estrogen production to retinal cell survival is limited. This study aimed to assess the role of locally synthesized estrogens in maintaining neuronal viability and glial stability in adult retinal explants.

Approach

An organotypic ex vivo retinal explant model was used, preserving the native laminar organization of the adult retina and physiological interactions between neurons and Müller glia. Explants were treated with letrozole (20 μM), a selective aromatase inhibitor, to block the conversion of androgens into estrogens and induce local estrogen deprivation. Untreated explants served as controls. After treatment, tissues were fixed and processed for immunohistochemistry. Apoptotic cell death was assessed using TUNEL labeling, while reactive gliosis was evaluated by analyzing glial fibrillary acidic protein (GFAP) expression.

Results

Letrozole-treated retinal explants showed a clear increase in TUNEL-positive cells compared with controls, indicating enhanced apoptotic cell death following inhibition of local estrogen synthesis. In parallel, a strong upregulation and spatial expansion of GFAP immunoreactivity was observed, consistent with Müller cell activation and reactive gliosis. These alterations were minimal in untreated explants, demonstrating that local estrogen deprivation is sufficient to disrupt retinal homeostasis and induce coordinated neuronal and glial dysfunction.

Conclusions

These findings provide direct functional evidence that local estrogen synthesis is required to maintain neuronal survival and glial homeostasis in the adult retina. Disruption of retinal steroidogenesis triggers key cellular features of retinal degeneration and supports local estrogen signaling as a potential therapeutic target in retinal neurodegenerative conditions.